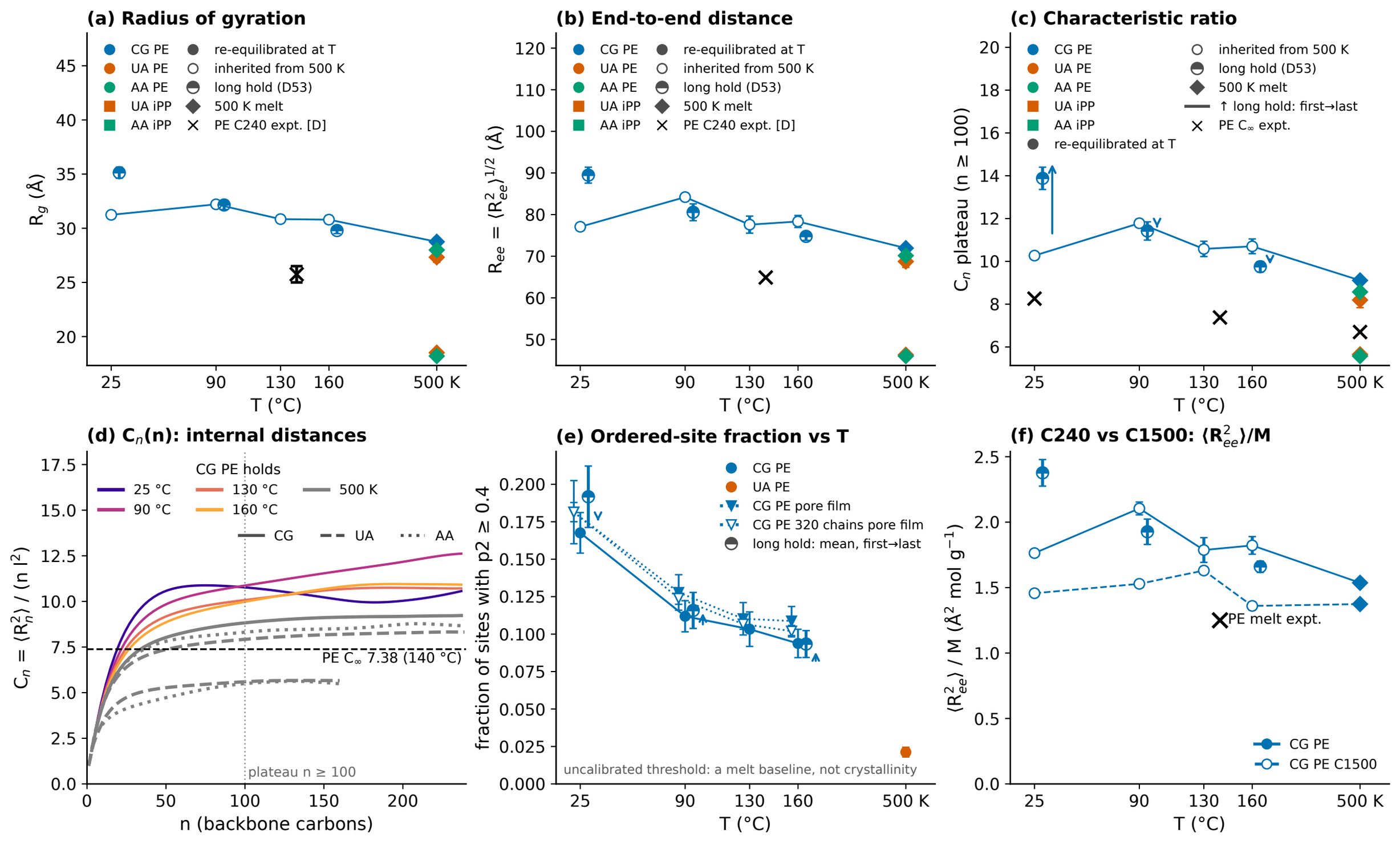

(a) Radius of gyration
CG PE
UA PE
AA PE
UA iPP
AA iPP
re-equilibrated at T
inherited from 500 K
long hold (D53)
500 K melt
PE C240 expt. [D]
R_g (Å)
T (°C)
(b) End-to-end distance
R_ee = ⟨R²_ee⟩^1/2 (Å)
(c) Characteristic ratio
C_n plateau (n ≥ 100)
↑ long hold: first→last
PE C∞ expt.
(d) C_n(n): internal distances
CG PE holds
25 °C
90 °C
130 °C
160 °C
500 K
CG
UA
AA
C_n = ⟨R²_n⟩ / (n l²)
PE C∞ 7.38 (140 °C)
plateau n ≥ 100
n (backbone carbons)
(e) Ordered-site fraction vs T
CG PE pore film
CG PE 320 chains pore film
long hold: mean, first→last
fraction of sites with p2 ≥ 0.4
uncalibrated threshold: a melt baseline, not crystallinity
(f) C240 vs C1500: ⟨R²_ee⟩/M
PE melt expt.
CG PE C1500
⟨R²_ee⟩ / M (Å² mol g⁻¹)